Response to Reviewer 4

State of Health Estimation and Target Leakage Auditing for Lithium Ion Batteries

Review dated 28 October 2026; revision prepared 1 November 2026. This revision uses the latest restored manuscript as the baseline, retains all 24 prior figures, and adds three source-based audit displays. Earlier drafts are preserved. We interpret the unnumbered title concern as R4.2 and the wording about "discount" as a request for five-fold membership and distribution reporting, not a 50% holdout.

The revision strengthens reproducibility and narrows claims. New sample-count, predictor-count and temperature-quota controls are supplementary diagnostics; they do not overwrite regression results. Missing original validation histories are disclosed, and no charging optimization experiments are claimed.

R4.1

Comment *The introduction retains "ExtraTrees reaching 99.1%" despite withdrawal in the abstract and conclusion.*

Response We checked the latest restored baseline: that phrase was already absent. We now add an explicit statement in the Introduction that 99.1% is withdrawn as a performance contribution and retained only as the motivation for an audit. The contribution paragraph describes observability, controlled leakage diagnostics and grouped/transfer evaluation. The only remaining 99.1% mentions identify the invalid earlier claim; none presents it as an achieved result.

Location Section I, second and final contribution paragraphs; Section V-A; Section X.

R4.2

Comment *The charging-optimization title is unsupported by charging-strategy experiments or revenue metrics.*

Response We chose the SOH/audit scope, rather than inventing charging experiments. The new title is "State of Health Estimation and Target Leakage Auditing for Lithium Ion Batteries." The abstract, contribution statement, prototype section and conclusion consistently identify current-cycle SOH estimation and leakage auditing. Charging recommendations and simulations remain exploratory software components, explicitly outside validated findings.

Location Title; Abstract; Sections I, VI, IX–X; Fig. 6.

R4.3

Comment *The V1/V2/V3 audit does not control feature family, sample size or temperature and cannot attribute MAE differences.*

Response We removed the "changing only the split" attribution and distinguish paired within-split leakage contrasts from cross-protocol comparisons. New ExtraTrees fits use (a) 1,152 training rows in every split, and (b) 836 training rows with fixed shared temperature quotas (416 at 4 °C, 380 at 24 °C, 40 at 43 °C). In each design all three contracts use identical train/test rows: observable 18, observable 18 plus four target proxies, and observable 18 plus four partition-wise shuffled proxies. These add 126 fits. Temperatures with zero common training support (22/44 °C) remain in test scoring. Temperature/C-rate support and error tables are exported. Different test populations and residual cell/history confounding remain; no causal split penalty is claimed.

Location Sections IV-A, VII-C and VIII–IX; Fig. 25; Supplement S3; reproducibility/controls and matched-control CSVs.

R4.4

Comment *Give exact definitions, windows and code generation for all four target proxies.*

Response Table 6 gives retention Q_k/Q_{ref}, cumulative capacity $\Sigma_{j\le k} Q_j$, first difference $Q_k - Q_{k-1}$ with zero for the first retained row, and a trailing mean of up to five SOH rows including the current row. Sorting, group resets, units and skipped-cycle behavior are explicit. Retention is exactly SOH/100 and is deliberately invalid. The supplement gives source paths and commands; tests verify boundaries and reset behavior. These variables are excluded from the observable benchmark and application inputs.

Location Section IV-A; Section VII-A, Table 6; Supplement S1; scripts/paper/r4_analysis.py and r4_test_analysis.py.

R4.5

Comment *Explain why 18 scalar features, 64 × 5 sequences and a 4.0–3.6 V window are physically reasonable, including temperature/rate dependence.*

Response We describe selection by observation availability and coverage of usage, voltage shape, current, partial energy and thermal response; all prespecified features were retained. Table 7 gives family counts, information and rationale, and the supplement supplies formulas. The fixed window harmonizes an early observed segment without using full capacity; it is not a demonstrated optimal or invariant SOC interval. Temperature/C-rate affect polarization and duration. The window is not adjusted by exposure or chemistry, and 64 bins are an engineering input resolution. Alternative windows and sampling resolutions were not tested; that limitation is explicit.

Location Section III-C; Section VII-A, Table 7; Section IX; Supplement S1 and exposure-support CSVs.

R4.6

Comment *List train/validation/test batteries and SOH distributions for every NASA fold and assess balance.*

Response The design is five folds, not a 50% holdout. Each seed-fold uses 19 training, five validation and six test batteries. We export 45 role inventories and all row-level role assignments for seeds 17, 42 and 2026. Table 8 gives all 15 test-fold class counts and SOH ranges; Supplement S2 lists every battery in all roles, SOH quartiles/ranges, and temperature/current support. The code asserts physical-cell disjointness and equality with the cached split identifiers. We explicitly state that GroupKFold does not stratify by SOH or temperature; missing EOL support is visible and balance is not claimed.

Location Sections IV-C and IV-G; Section VII-B, Table 8; Supplement S2; nasa_split_inventory.csv and nasa_split_records.csv.

R4.7

Comment *Specify whether frozen preprocessing includes scaling, missing interpolation and temperature encoding.*

Response The frozen quantities are training-fitted scalar medians, the StandardScaler used by SVR/Ridge/KNN, sequence-channel medians/means/scales, neural target normalization and fitted ensemble parameters. Tree pipelines do not standardize. Within-record interpolation is a fixed operation confined to the completed observed partial segment, not fitted on a pool of target records. Ambient temperature is numeric Celsius and observation status binary; no target-fitted categorical encoding occurs. Missing thermal channels use the fixed ambient fallback and observation flag. Oxford adapter rules and reconstructed current are distinguished from fitted normalization.

Location Sections III-C, IV-D and IV-G; Supplement S1.

R4.8

Comment *Clarify battery-macro MAE versus sample MAE after averaging repeated predictions.*

Response Both metrics are now defined by explicit formulas using the same per-cycle mean of the three held-out predictions. Sample MAE weights each retained cycle equally; battery-macro MAE weights each battery equally. The mean of 15 fold MAEs is separately labeled and is not substituted for either metric. Table 3 remains sample MAE; Table 4 labels macro/sample columns. Supplement S5 supplies all three for all 20 models and the fold-level records. Post-evaluation rankings are described as descriptive rather than unbiased deployment-model selection.

Location Section IV-G; Sections V-B and VII; Tables 3–4; Fig. 9; Supplement S5.

R4.9

Comment *Explain validation/test cell separation and whether ensemble weights are re-estimated outside each fold.*

Response Training, validation and test cells are mutually disjoint within each seed-fold. Base models train on training cells; inverse-validation-MAE weights or the Ridge meta-model fit validation predictions and are frozen for that fold's test cells. Each outer fold/seed gets its own fit. Full-source zero-shot ensembles use NASA out-of-fold predictions for meta fitting and NASA-only base refitting before external scoring. The secondary residual calibration uses validation-battery cross-fitting, avoiding calibration on in-sample meta outputs. Split identities and per-fold secondary weights are exported; tests independently check separation.

Location Sections IV-F–G; Supplement S2/S4; secondary_ensemble_fold_weights.csv; split JSON records.

R4.10

Comment *Report actual stopping epochs and validation/test gaps for every deep model.*

Response We recovered and export all 420 original grouped deep-fit records, with dataset/model/seed/fold, executed epochs and training/test MAE. Table 9 reports the NASA ranges and gaps; Supplement S4 reports all three datasets. Executed epochs include the patience tail and must not be called restored-best epochs. Two original vanilla-LSTM fits reached the 200-epoch cap. The original artifacts do not retain recoverable per-epoch validation histories or best-epoch identifiers. We therefore do not fabricate original validation gaps: we separately report validation/test MAE and differences for all 150 saved NASA secondary reruns (Fig. 27). The monitored objectives are now stated correctly, including combined VAE loss and weighted two-head TensorFlow loss. This is partial recovery of the original audit trail, with the remaining provenance limit explicit.

Location Section IV-B; Section VII-B/E, Table 9 and Fig. 27; Section IX; Supplement S4; original_deep_fold_stopping.csv and secondary_nasa_validation_test_gaps.csv.

R4.11

Comment *Revise inference with four CALCE cells; specify the Wilcoxon unit and multiple-comparison family.*

Response The signed-rank unit is one paired per-battery MAE after prediction averaging (30 NASA cells), not a fold or a seed-cell replicate. The two-sided, zsplit Wilcoxon family comprises exactly 19 comparisons with Weighted Average Ensemble, adjusted by Holm. The reference was identified from the same benchmark, so inference is exploratory and does not correct selection. Bootstrap intervals resample batteries with replacement and condition on fitted predictions; they do not account for retraining or provide new cells. We retain the wide CALCE interval and add leave-one-observed-cell-out influence diagnostics (sample MAE 0.968–1.345 pp). With four cells, neither stable population endpoints nor model superiority is asserted.

Location Sections IV-C, V-F and IX; Fig. 23; Supplement S6; verified Wilcoxon and leave-one-cell-out CSVs.

R4.12

Comment *The small improvement without temperature needs permutation-importance intervals or interaction analysis.*

Response We added ten-repeat individual and block permutations for five grouped fits, with pointwise 95% intervals from 10,000 paired battery-cluster draws. Both pooled and within-battery permutations are reported, plus $\Delta(T,V)-\Delta(T)-\Delta(V)$ to diagnose nonadditivity. Within-cell shuffling preserves ambient exposure; its thermal-block interval includes zero. The paired no-temperature retraining sample-MAE change is −0.130 pp (−0.352 to 0.133), also spanning zero. These are conditional predictive diagnostics, not causal effects or family-wise interaction tests. Original fold-mean ablation results remain intact and are distinguished from the new pooled sample metric.

Location Sections V-E and VII-D; Fig. 26; Supplement S7; permutation_cluster_intervals.csv and temperature_ablation_paired_interval.csv.

R4.13

Comment *Consider the five suggested 2026 hybrid-model papers.*

Response We checked publisher records for all five. We added Chen's energy-demand XGBoost/deep-learning paper as limited methodological context [26], while stating that its forecasting endpoint cannot validate battery SOH or charging control. The consumer-click, trade-fraud, export-volume and inflation papers concern different endpoints and are not added as battery evidence. This is a relevance decision, not a judgment of their quality. Publisher metadata for the export paper lists Wenhao Wang, Siyi Shen and Ye Yuan; the supplied citation's third author differs. The reference assessment below identifies every paper and decision.

Location Section I, related methodological-context paragraph; Reference [26]; Supplement S8; reproducibility/reference_assessment.json.

Assessment of suggested references

Haotian Zhou. Research on the Fusion Model of DeepFM and XGBoost for Digital Consumer Behavior Prediction. DOI 10.71451/ISTAER2617. discussed in response, not added to manuscript. Consumer click prediction concerns a different endpoint and validation unit; it does not substantiate battery SOH validity or charging optimization. Publisher: https://istaer.online/index.php/Home/article/view/No.2617

Xi Zeng. Cross-Border Trade Fraud Detection via Integrated Heterogeneous Graph Neural Network and XGBoost. DOI 10.71451/ISTAER2603. discussed in response, not added to manuscript. Fraud classification does not address battery observability, held-out cells or charging control. Publisher: https://istaer.online/index.php/Home/article/view/No.2603

Guona Chen. Research on an Energy Economic Demand Time Series Prediction Model Based on the Integration of XGBoost and Deep Learning. DOI 10.67541/jdc2601. added as limited methodological context, reference 26. Energy-demand forecasting uses tree/deep fusion; the manuscript explicitly distinguishes this forecast endpoint from retrospective battery SOH estimation. No numerical results transferred. Publisher: https://dipscie.com/index.php/JDC/article/view/10.67541_jdc2601

Wenhao Wang, Siyi Shen, Ye Yuan. Forecasting Short-Term Export Volumes with Hybrid Models Integrating SARIMA with Attention-Based LSTM. DOI 10.71451/ISTAER2601. discussed in response, not added to manuscript. Export forecasting is outside the battery estimation endpoint. Publisher lists Yuan Y as third author, whereas the review lists Wang Y. Publisher: https://istaer.online/index.php/Home/article/view/No.2601

Guona Chen. Research on Algorithm Improvement of ARIMA-LSTM Hybrid Model in Time Series Prediction of Inflation Rate. DOI 10.71451/ISTAER2605. discussed in response, not added to manuscript. Inflation time-series forecasting does not provide battery-specific estimation or validation evidence. Publisher: https://istaer.online/index.php/Home/article/view/No.2605